Sean Luka Girgis

Senior Python Backend / Data Engineer | APIs, Production Systems, PySpark & Microservices

214-315-2190 | seanlgirgis@gmail.com

LinkedIn | GitHub | Portfolio

Professional Summary

Senior Python backend/data engineering professional with 20+ years of enterprise technology experience across Python, SQL, API development and consumption, production system support, distributed data workflows, PySpark/Spark, microservices concepts, troubleshooting, documentation, and stakeholder collaboration. Strong background building and supporting reliable Python-driven workflows, integrating with internal systems and services, validating production outputs, debugging complex enterprise issues, and improving reliability of business-critical platforms. Best aligned to backend/data engineering roles requiring Python, API integration, production support, microservices architecture concepts, distributed systems, PySpark/Databricks exposure, and fast ramp-up into complex platforms; pricing-engine domain and pure application microservice ownership are positioned as transferable rather than overstated prior ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built and supported Python/Pandas/PySpark and SQL workflows ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for analytics, reporting, forecasting, validation, and operational decision support.
- Developed Python-based data and platform workflows that integrated multiple internal systems, monitoring feeds, reporting layers, and warehouse-backed services into reliable business-facing outputs.
- Consumed and integrated enterprise platform data from systems including BMC TrueSight, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and infrastructure telemetry sources.
- Supported production systems by tracing failed outputs, source-feed behavior, Python processing, SQL transformations, API-style data integration issues, latency symptoms, and downstream reporting discrepancies.
- Created validation and regression-style checks for row counts, nulls, duplicates, freshness, abnormal deltas, unmatched records, and output comparisons against trusted prior runs.
- Used AWS S3, Glue, Athena, Redshift, EC2/ECS concepts, PySpark, and SQL to support scalable processing, cloud-oriented data workflows, and production reporting reliability.
- Created runbooks, data flow notes, API/data integration notes, metric definitions, troubleshooting guides, and support procedures to improve maintainability and handoff.
- Partnered with application, infrastructure, analytics, engineering, and business stakeholders to clarify requirements, prioritize fixes, explain system behavior, and deliver improvements under production pressure.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Supported production-facing Brand.com and hospitality platforms using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.

- Analyzed application behavior, transaction patterns, monitoring signals, and production-like symptoms to identify reliability and performance issues.
- Created dashboards and reporting views that converted raw monitoring and transaction data into actionable performance, reliability, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical platform improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services application monitoring environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Provided production troubleshooting guidance to application, middleware, infrastructure, and operations teams using CA APM/Introscope telemetry, dashboards, alerts, and transaction visibility.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable operational processes.
- Created technical documentation, support procedures, onboarding material, dashboard standards, alert thresholds, and troubleshooting notes used by multiple teams.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and production-readiness testing.
- Documented test results, expected behavior, defects, risks, performance metrics, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led validation and production-readiness support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported phased cutover, troubleshooting, documentation, and defect resolution for a mission-critical enterprise transaction platform.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, VC++ components, and DB2-backed integration modules in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using C, C++, Pro*C, Oracle, PL/SQL, SQL scripts, XML, Java/J2EE concepts, and enterprise messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, troubleshooting, and release support for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Flagship Projects

ServiceCall AI RAG Demo
Built staged Python proof-of-concepts with FastAPI service layers, Pydantic request/response contracts, Docker-first smoke tests, deterministic API endpoints, and testable service behavior.
Technologies: Python, FastAPI, Docker, Pydantic, pytest, REST API Concepts

Production Data Quality and RCA Workflow
Built validation and troubleshooting workflows for production reporting datasets, including row-count checks, missing feed diagnosis, unmatched join checks, prior-run comparisons, and documented recovery steps.
Technologies: Python, SQL, Pandas, Oracle, Data Validation, RCA

Serverless Lakehouse Platform (AWS)
Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet with validation checkpoints, SQL quality checks, and curated analytics datasets.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL

High-Throughput Oracle RAC Migration
Led validation and production-readiness support for migration from 200+ MySQL nodes to a 6-node Oracle RAC platform, using C++/OCCI/OCI validation utilities and SQL optimization to preserve transaction correctness and performance.
Technologies: C++, OCCI, OCI, Oracle RAC, MySQL, SQL